

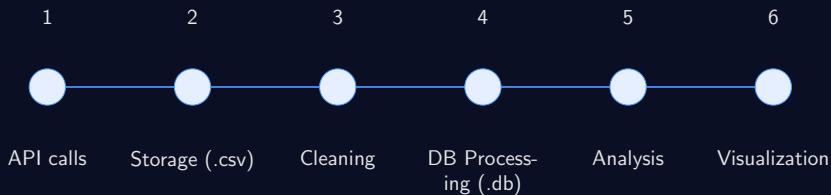
The idea?

The idea?

Collect public available data published by the Metropolitan Transportation Authority (MTA), store it and visualize it.

The Objective

Build an end-to-end data pipeline that pulls real MTA subway data, cleans and stores it in a local database, analyzes patterns and trends, and visualizes the results in a dashboard.



Cronological setup

1. API calls, data issues, estimated times, scripts.

On this stage the objective is to collect the publicly available data. A Google search revealed this dataset:

MTA Subway Hourly Ridership: 2020-2024

Category	Information
Organization	New York City Transit, Staten Island Railway, Roosevelt Island Operating Corporation
Time Period	2020 – 2024
Posting Frequency	Static - Not Updated
Dataset Owner	Metropolitan Transportation Authority
Contact Information	OpenData@mtahq.org
Coverage	New York City
Granularity	Hour, mode, subway station complex, fare medium, class of fare payment

What's in this dataset?

1. Dataset overview

What's in this dataset?

We have 121M rows, 12 columns.

Column name	Description
transit_timestamp	Timestamp payment took place in local time.
transit_mode	Distinguishes between the subway, Staten Island Railway, and the Roosevelt Island Tram
station_complex_id	unique identifier for station complexes
ridership	Total number of riders that entered a subway complex via OMNY or MetroCard at the specific hour and for that specific fare type.
fare_class_category	The class of fare payment used for the trip.
borough	Represents one of the boroughs of New York City serviced by the subway system
payment_method	Whether the payment method used to enter was from OMNY or MetroCard.

1. API calls, data issues, estimated times, scripts.

Figure: Urbain Le Verrier

Figure: John Couch Adams

The stars of the show II

Figure: Alexis Bouvard

Figure: Johann Gottfried Galle

Galle found Neptune in the same night he received Le Verrier's letter, within 1° of the predicted position.

The stars of the show II

Figure: Alexis Bouvard

Figure: Johann Gottfried Galle

Galle found Neptune in the same night he received Le Verrier's letter, within 1° of the predicted position.

Motivation: Uranus had residuals I

Figure: Table showing the errors in longitude for the predicted orbit of Uranus

Observed positions of Uranus deviated from predictions using only known planets (Bouvard's tables). Bouvard got the bright idea (probably from Laplace) that perhaps the two large planets, Jupiter and Saturn, were “perturbing” the orbit of Uranus

Perturbation theory I

Perturbation theory is a mathematical approach that approximates solutions to problems that cannot be solved analytically, starting from the exact solution of a simpler related case.

Base problem (unperturbed orbit): two bodies (Sun + Planet U)

(1) Gravitational Force: $\mathbf{F} = -G \frac{M_{\odot} m_2}{r^2} \hat{\mathbf{r}}$

(2) Newton's Second Law: $\mathbf{F} = m_2 \ddot{\mathbf{r}}$

Combine: $m_2 \ddot{\mathbf{r}} = -G \frac{M_{\odot} m_2}{r^2} \frac{\mathbf{r}}{r}$

$$\Rightarrow \boxed{\ddot{\mathbf{r}} = -\frac{GM_{\odot}}{r^3} \mathbf{r}}$$

Solution: Keplerian ellipse with parameters $(a, e, i, \omega, \Omega, M)$.

Perturbation theory II

Perturbed problem (adding a third body): planet P (mass m' at r')

$$\ddot{\mathbf{r}} = -\frac{GM}{r^3}\mathbf{r} + \epsilon \mathbf{a}_{\text{pert}}, \quad \mathbf{a}_{\text{pert}} = -Gm' \left(\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} + \frac{\mathbf{r}'}{r'^3} \right)$$

where $\epsilon \ll 1$ and $\mathbf{a}_{\text{pert}}$ represents the gravitational influence of another planet.

Expanding orbital elements as:

$$a(t) = a_0 + \epsilon a_1(t), \quad e(t) = e_0 + \epsilon e_1(t), \quad \dots$$

Substituting and keeping first-order terms in ϵ :

$$\frac{d\omega}{dt} = \frac{3}{4}n \frac{m'}{M_{\odot}} \left(\frac{a}{a'} \right)^3 \frac{1}{(1 - e'^2)^{3/2}}$$

This gives the secular precession of the perihelion, i.e., the slow change in the orientation of a celestial body's orbit, primarily caused by gravitational interactions with other bodies in the system.

Interpretation:

Last equation can be just expressed as

$$\frac{d\omega}{dt} = C n \epsilon \left(\frac{a}{a'} \right)^3 + \mathcal{O}(\epsilon^2)$$

This means that a distant planet with small mass m' produces a slow secular precession of Uranus' perihelion (or a slow drift in mean longitude) proportional to m' and $(a/a')^3$. Over decades this tiny rate integrates to an observable angular residual.

Physical meaning of the previous equation

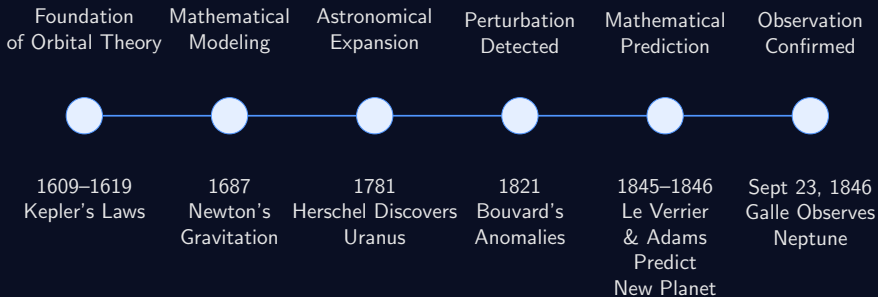
Source: Wikipedia

Direct perturbations involve computing the effect of a known body, such as Jupiter's influence on Uranus. However, Uranus's orbit showed unexplained deviations.

Le Verrier and Adams deduced that these were caused by an unknown planet—Neptune.

Inverse perturbation: the effect is observed but the cause (mass and position) must be inferred. Is much more complex than the direct case.

Timeline



Chronological roadmap

I integrated a toy solar system in Python with the Sun, Jupiter, Saturn, Uranus;

(A) without Neptune, (B) adding Neptune.

I plotted Uranus' mean longitude difference vs time and show the secular drift produced by Neptune—the effect Le Verrier (and later Adams) inferred.

Simplified Coplanar Orbits

Simplified Coplanar Orbits